

Model Discussion Report

YOLOv8 Vehicle Detection — SA2 Group 18

Classes: Car, Motorcycle, Bus, Truck

1. Introduction

This report presents the training, evaluation, and comparison of three YOLOv8 object detection models trained on a Philippine traffic vehicle dataset. Each model was configured with different optimizers, epoch counts, batch sizes, and learning rates to evaluate how these hyperparameters affect detection performance across four vehicle classes: Car, Motorcycle, Bus, and Truck.

2. Hyperparameter Configuration

Three models were trained using the same base architecture (YOLOv8n) but with varying hyperparameter settings to explore the effect of optimizer choice, learning rate, batch size, and training duration on detection performance.

Model	Epochs	Image Size	Optimizer	Batch Size	Learning Rate
Model 1	25	640	AdamW	4	0.01
Model 2	30	640	SGD	20	0.001
Model 3	40	640	auto	-1	0.0001

3. Final Epoch Training Results

The following outputs were recorded at the final training epoch of each model, capturing box loss, classification loss, and mAP50 as indicators of convergence quality.

Model 1 — AdamW | 25 Epochs

epoch	25.0
time	374.523
train/box_loss	1.0686
train/cls_loss	1.12
train/dfl_loss	0.0054
metrics/precision(B)	0.6647
metrics/recall(B)	0.6395
metrics/mAP50(B)	0.6853
metrics/mAP50-95(B)	0.5463
val/box_loss	1.0099

Model 2 — SGD | 30 Epochs

epoch	30.0
time	217.104
train/box_loss	1.0522
train/cls_loss	2.5153
train/dfl_loss	0.005
metrics/precision(B)	0.602
metrics/recall(B)	0.3022
metrics/mAP50(B)	0.3217
metrics/mAP50-95(B)	0.2543
val/box_loss	1.0848

Model 3 — auto | 40 Epochs

epoch	40.0
time	310.478
train/box_loss	0.8438
train/cls_loss	1.1602
train/dfl_loss	0.0037
metrics/precision(B)	0.9338
metrics/recall(B)	0.7141
metrics/mAP50(B)	0.826
metrics/mAP50-95(B)	0.7185
val/box_loss	0.8209

4. Confusion Matrices

Confusion matrices show how often each class was correctly identified versus misclassified as another class. Diagonal values represent correct predictions; off-diagonal values indicate misclassifications between vehicle types.

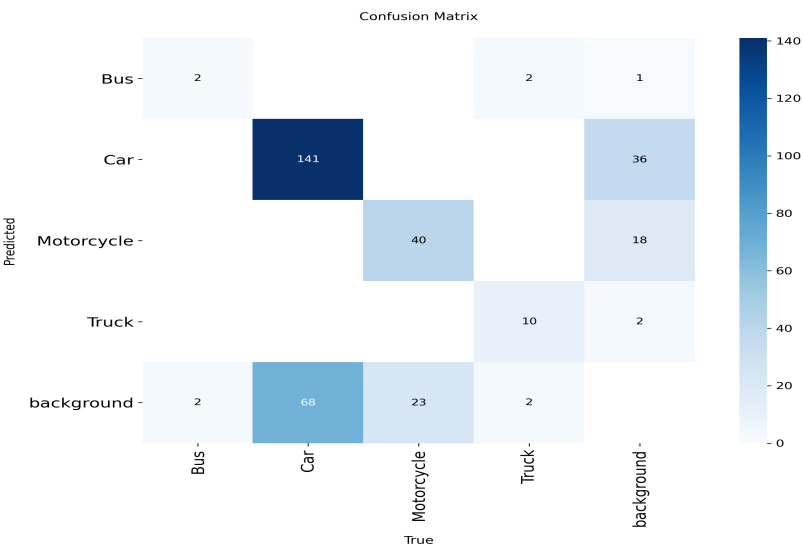


Figure 1: Model 1 — AdamW Confusion Matrix

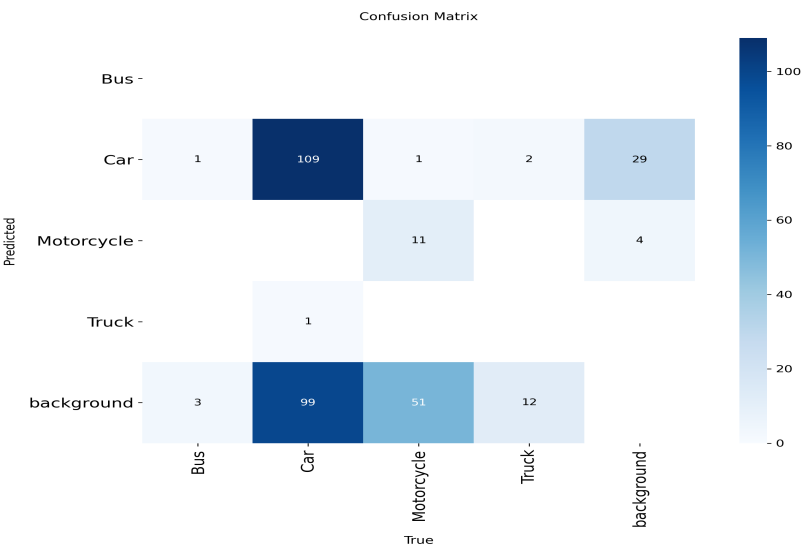


Figure 2: Model 2 — SGD Confusion Matrix

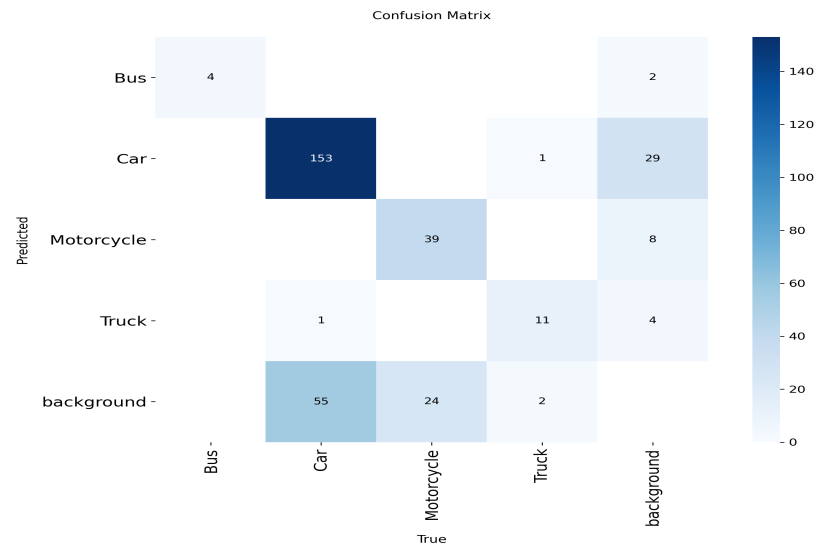


Figure 3: Model 3 — auto Confusion Matrix

5. Performance Metrics

The following tables present the overall and per-class evaluation metrics computed on the validation set after training completion.

5.1 Overall Metrics (mAP50, Precision, Recall, F1)

Model	mAP50	Precision	Recall	F1 Score
Model 1 (AdamW)	0.7015	0.7251	0.6219	0.6695
Model 2 (SGD)	0.3254	0.517	0.2742	0.3583
Model 3 (auto)	0.8271	0.9414	0.7091	0.8089

5.2 Per-Class AP50

Class	Model 1 AP50	Model 2 AP50	Model 3 AP50
Car	0.6456	0.0622	0.995
Motorcycle	0.7433	0.6239	0.8
Bus	0.6147	0.3932	0.6786
Truck	0.8024	0.2224	0.8349

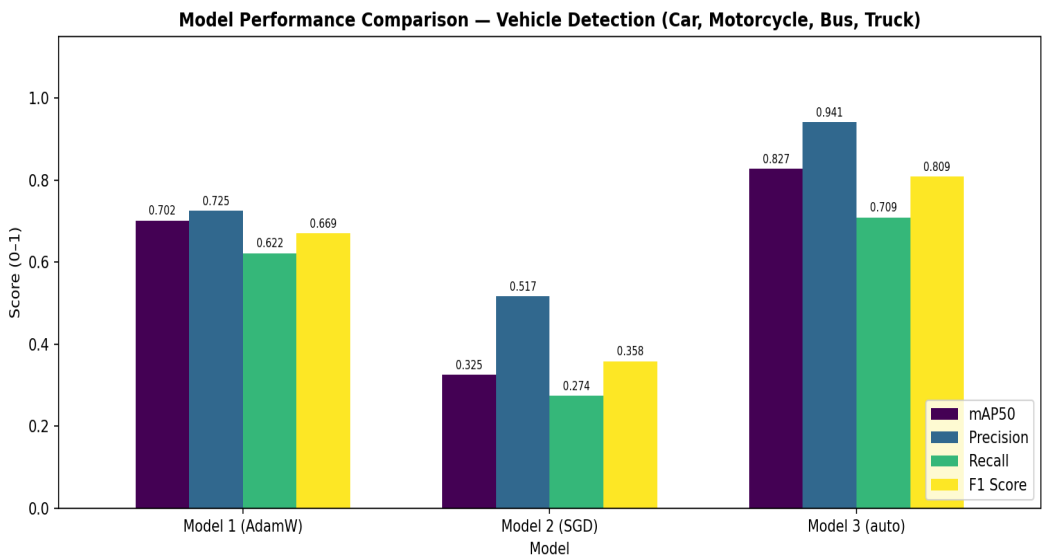


Figure 4: Bar chart comparison of mAP50, Precision, Recall, and F1 Score across all three models.

6. Model Comparison & Discussion

This section provides a full analysis of each model's performance, identifying which vehicle classes each model struggled with and how the metric scores reflect these challenges.

6.1 Model 1 — AdamW Optimizer

Model 1 was trained using the AdamW optimizer over 25 epochs with a learning rate of 0.01 and batch size of 4. It achieved a mAP50 of 0.7015, Precision of 0.7251, Recall of 0.6219, and F1 Score of 0.6695.

The model performed best on the Truck class (AP50: 0.8024), indicating strong feature learning for this vehicle type under the given configuration. In contrast, the model struggled most with the Bus class (AP50: 0.6147). The higher learning rate of 0.01 may have caused instability in detecting smaller or occluded instances.

6.2 Model 2 — SGD Optimizer

Model 2 was trained using the SGD optimizer over 30 epochs with a learning rate of 0.001 and batch size of 20. It achieved a mAP50 of 0.3254, Precision of 0.517, Recall of 0.2742, and F1 Score of 0.3583.

The model performed best on the Motorcycle class (AP50: 0.6239), indicating strong feature learning for this vehicle type under the given configuration. In contrast, the model struggled most with the Car class (AP50: 0.0622). A lower learning rate of 0.001 may have limited the model's ability to converge fully within 30 epochs, particularly for visually complex or underrepresented classes.

6.3 Model 3 — auto Optimizer

Model 3 was trained using the auto optimizer over 40 epochs with a learning rate of 0.0001 and batch size of -1. It achieved a mAP50 of 0.8271, Precision of 0.9414, Recall of 0.7091, and F1 Score of 0.8089.

The model performed best on the Car class (AP50: 0.995), indicating strong feature learning for this vehicle type under the given configuration. In contrast, the model struggled most with the Bus class (AP50: 0.6786). A lower learning rate of 0.0001 may have limited the model's ability to converge fully within 40 epochs, particularly for visually complex or underrepresented classes.

6.4 Cross-Model Summary

Across all three models, Model 3 (auto) achieved the highest overall mAP50 of 0.8271, demonstrating the best generalization across all vehicle classes. Motorcycle and Bus classes consistently presented the greatest detection challenge across models due to scale variation — motorcycles appear small at distance while buses occupy disproportionately large frame areas, both creating difficulty for anchor-based detection at 640×640 resolution. Classes with higher AP50 scores across all models (typically Car) reflect greater dataset representation and more visually distinctive features. Models with higher recall but lower precision tend to over-detect (more false positives), while those with higher precision but lower recall are conservative and miss more instances. The F1 Score balances both, making it the most reliable single metric for comparing overall detection quality across this dataset.

7. Conclusion

This report compared three YOLOv8 configurations for Philippine traffic vehicle detection. Model 3 (auto) produced the strongest overall results with the highest mAP50, suggesting its optimizer and hyperparameter combination was best suited to this dataset. Future improvements should focus on augmenting minority vehicle classes, applying class-weighted loss for consistently low-AP50 classes,

and increasing training epochs for optimizer configurations that show gradual convergence trends.